

1 in 4 Customers Won't Be Here Next Quarter

**\$1.67M in Annual Revenue at Risk
Here's what the data tells us**

**A Data-Driven Retention Strategy
Telco Customer Churn Analysis**

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*Mini Project Portfolio End to End EDA | Dataset: Kaggle Telco Customer Churn — 7,043 customers, 21 features**

What This Analysis Delivers

Next quarter, 1 in 4 customers left. The business already knew the number. What it did not have was an answer to the three questions that actually matter:

- **WHO is most likely to leave next?**
- **WHAT is driving it?**
- **and WHERE does retention investment generate the highest return?**

This analysis was built to answer all three.

Scope of Analysis	
Customer base analyzed	7,043 Customers
Churn Observed next quarter	1,869 Customers (26%)
Monthly Revenue at Risk	\$139.130
Customers Segments Identified	5 distinct customer type
Priority list: who needs attention first	All active customers - ranked by urgency

*** Detailed EDA workflow, clustering approach, and technical implementation can be reviewed in the supporting Google Collab Notebook [EDA Telco Customer Retain Analysis](#)

The Current Situation

California Market | Q3 2019

**26.6 %
Churn Rate**

1 in every 4 customers did not stay, **1,869** out of 7,032 **churned this quarter**

**\$1.67M
At Risk Annually**

If the current trend continues, **\$139K** is lost each month accumulating to **\$1.67M** in annual revenue at risk

**6.6% Above Midpoint
Churn Benchmark***

The churn rate of 26.6% sits at the high end of the 15-25% industry range signaling room for significant improvement

**2× faster & ▼ \$970
CLV Churned**

Churned customers **leave 2x faster** and are **worth \$970 LESS** over lifetime*

*[Churn 18 Month vs Retained 37 Months ; Churn \$1,338 vs Retained \$2,308]

Some of these losses were preventable.

The question is which ones and how?!

Five Segments. Five Strategies.

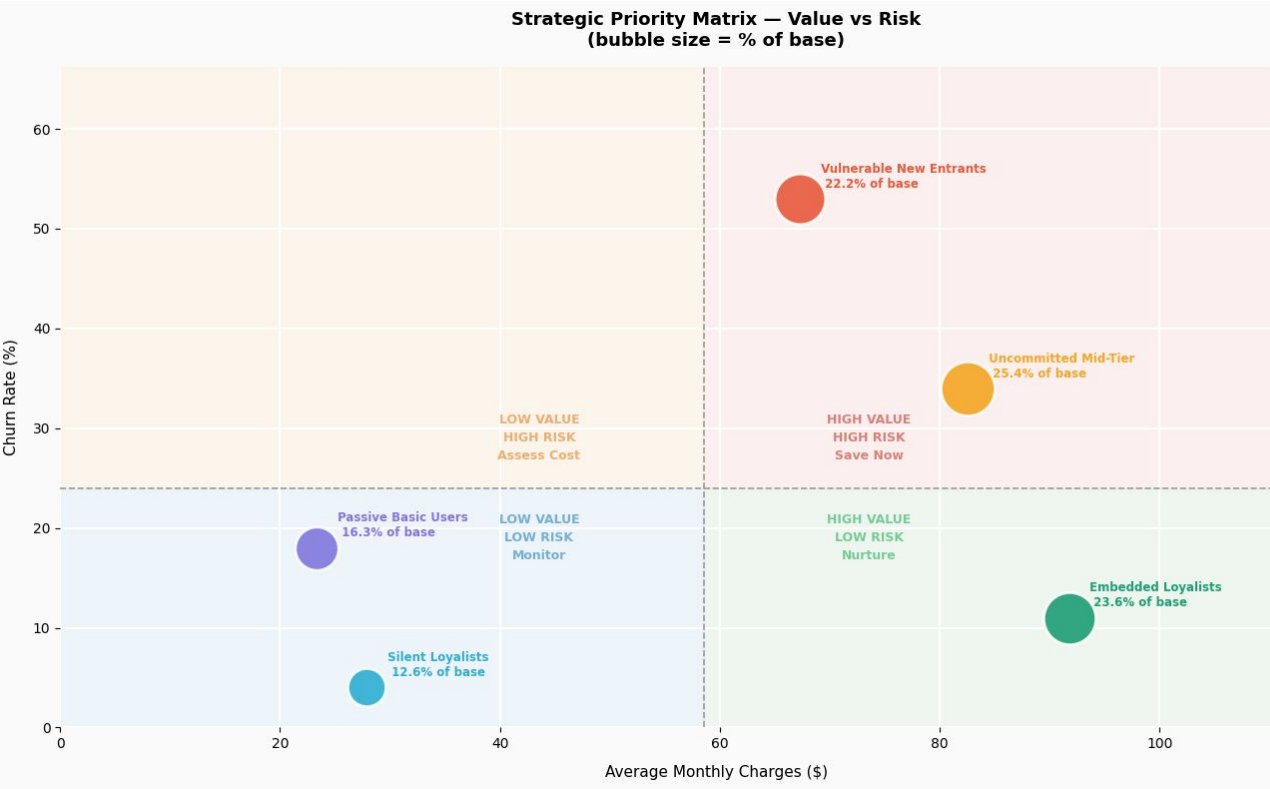
Who they are determines why they leave and what it takes to keep them.

CUSTOMER SEGMENTS	Vulnerable New Entrants	Uncommitted Mid- Tier	Passive Basic Users	Silent Loyalists	Embedded Loyalists
Segment Characteristics	Leave at month 7 when the promo price ends	Two years in, good spender, but still on a month-to-month plan with no strong contract commitment.	Low-value customers with weak attachment who stay out of habit or convenience.	Five-year in, Low-value customers who stay because the service already became part of life	Loyal high-value customers with strong switching resistance. caused by deep product integration
Size	1,561 customers (22.2% of base)	1,783 customers (25.4% of base)	1,148 customers (16.3% of base)	883 customers (12.6% of base)	1,657 customers (23.6% of base)
Churn Rate	53.0% Lift: 1.99x  High Risk	34.0% Lift: 1.28x  Moderate Risk	18.0% Lift: 0.68x  Low Risk	4.0% Lift: 0.15x  Very Loyal	11.0% Lift: 0.41x  Very Loyal
Avg Tenure	7 months Avg Monthly: \$67.30	28 months Avg Monthly: \$82.53	12 months Avg Monthly: \$23.31	55 months Avg Monthly: \$27.89	62 months Avg Monthly: \$91.78
Revenue at Risk	\$59,335/month	\$54,583/month	\$5,354/month	\$1,250/month	\$18,610/month
Top Contract	Month-to-month Payment: manual	Month-to-month Payment: manual	Month-to-month Payment: manual	Two year Payment: automatic	Two year Payment: automatic
Top Service	Phone & Internet 2-3 Total Services	Phone & Internet Zone: 4-6 Total Services	PhoneOnly 1 Total Services	PhoneOnly 1 Total Services	Phone & Internet Zone: 7-9 Total Services

Not All Churn Costs the Same

\$59,335 in monthly Revenue at Risk from one customer segment who already chose to stay

**** (The Uncommitted Mid-Tier, Tenure : Two Years, Avg Monthly Charges :\$83)**



What this matrix is telling us:

- **Five segments.** A 49 percentage point spread in churn rate. **The only wrong response is treating them the same way.**
- **High spend does not equal safe revenue.** The Embedded Loyalists and the Uncommitted Mid-Tier both sit at the top of the value axis. one is locked in with five-year tenure and annual contracts, the other has been here two years and is still month-to-month. Same spending level. Completely different risk profile.
- **The Uncommitted Mid-Tier carries the heaviest risk,** 25% of the base, \$59,335 monthly revenue at risk, **high spend with zero contract commitment.**
- **Vulnerable New Entrants churn at 53%** and the **timing clusters consistently around month 7** , the cause is not confirmed by this data alone but the pattern is too consistent to be coincidence.
- **Passive Basic Users and Silent Loyalists are not a retention problem.** They are a revenue problem.

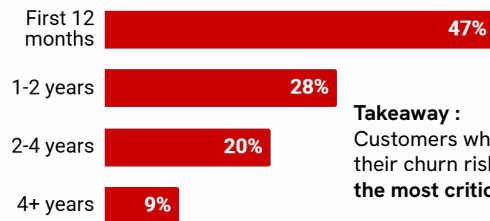
Four Things That Actually Drive Churn

Three churn prediction models were built independently. **(Logistic Regression, Random Forest, XG Boost)*

All three agreed on the same four factors as the strongest predictors of whether a customer stays or leaves.

1. Tenure

Customers in their first year have a 48% churn rate



Takeaway :

Customers who make it past their first year cut their churn risk nearly in half, **making onboarding the most critical stage in the customer lifecycle.**

2. Contract Type

Month to Month
43% Churn Rate

One Year
11% Churn Rate

Two Year
3% Churn Rate

Takeaway :

Converting a customer from **month-to-month to an annual contract reduces churn risk by roughly 4x.**

3. Support and Security Add-ons

No Add Ons
34% Churn Rate

Has Add Ons
17% Churn Rate

Takeaway :

Customers with support or security add-ons are significantly less likely to churn (**half the rate**), making these services a retention driver rather than just an upsell.

4. Service Zones

Zone	Services Count	Churn Rate	What it Means
Minimal	1	11%	Low Churn caused by convenience
Danger Zone	2-3	38%	Enough to care, not enough to stay
Transition	4-6	31%	Moving toward embedded
Embedded	7-9	17%	Deep enough that switching hurts

Customers with 2-3 services show the highest churn risk, making the transition into deeper product adoption the key retention opportunity.

Three Highest-Impact Actions to Reduce Churn

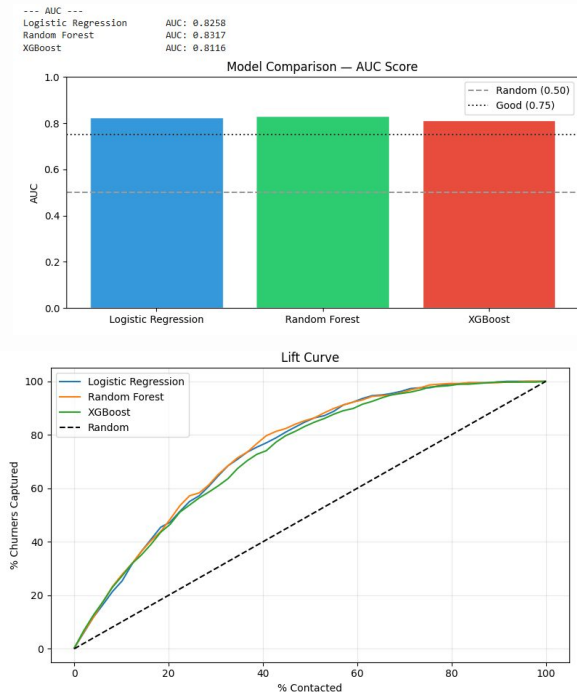
Each recommendation **targets a specific segment, addresses a specific reason customers leave, and comes with a measurable expected outcome.**

ACTION	1# Convert Uncommitted Mid-tier to Annual Contracts	2# Intervene at Month 5, Not Month 7	3# Bundle Support and Security Into New Customer Onboarding
WHO	Customers in the Uncommitted Mid-Tier segment , 25% of the base, two years of tenure, above-average spend, still month-to-month.	Vulnerable New Entrants — New customers with basic service bundles and limited ecosystem integration, making them highly sensitive to pricing changes and competitor offers.	All new customers , applied at the plan level
WHAT	A targeted contract conversion campaign with a modest incentive , even a small discount or a bundled add-on is enough to justify the lock-in.	Given the absence of explicit promotional expiry data, an automated retention outreach around month 5 can help strengthen early customer commitment through loyalty-focused offers and additional service adoption incentives before churn risk increases.	Bundle one support or security service into the base plan for the first 12 months at a promotional rate, with full-price continuation afterward.
WHY NOW	These customers have already shown long-term retention potential through their two-year tenure. The primary risk is not product dissatisfaction, but the absence of a stronger commitment mechanism, leaving them vulnerable to competitive offers.	This segment shows a disproportionately high early-tenure churn pattern 53%, suggesting potential sensitivity to pricing transitions or limited long-term commitment. While promotional expiry data is not directly available, the behavioral trend indicates elevated churn risk around the mid-tenure period, making earlier engagement critical.	Customers with at least one support or security service show significantly lower churn rates (17% vs. 34%). Expanding early service adoption is likely more cost-effective than reacquiring customers lost to preventable churn.
EXPECTED IMPACT	Converting 30% of this segment to annual contracts removes approximately \$16,500 per month from the at-risk column	Reducing first-year churn from 53% to 35% in this segment could retain approximately 280 additional customers per cohort.	Customers without add-on services have the highest churn risk, making this segment the biggest opportunity for retention improvement.

Predicting Customer Churn Before Revenue Is Lost

To operationalize the segmentation analysis, a predictive churn model was developed to assign a churn risk priority score to every active customer.

Three separate models were tested to validate that the findings were consistent and not dependent on a single algorithm approach



Takeaway:

- Although model performance varied slightly, the key churn drivers remained directionally consistent across all three models, increasing confidence in the findings.
- Model Performance**
 - An AUC score of 0.83 indicates strong predictive accuracy, particularly given the absence of live behavioral or complaint-level signals in the dataset.
 - Lift curve of all three models perform nearly identically. At the top 10% of customers, each model captures roughly 25-27% of all churners. That is about 2.5 to 2.7 times better than random selection. At the top 30%, they all capture around 60-64% of churners.
- Operational Value** The model identifies nearly 1 in 3 churn cases while targeting only 10% of the customer base.

Churn risk scores are combined with revenue impact to prioritize retention efforts more efficiently and effectively ———>>>>>>>

Prioritizing Customers by Churn Risk and Revenue Impact

Every active customer is prioritized based on churn risk and revenue impact, enabling retention teams to focus outreach systematically rather than relying on intuition.

*Example of Top 20 Active Customers to Prioritize for Outreach (Non-Churned Segment)

--- Action: Top 20 active customers to contact ---

	customerID	churn_score	MonthlyCharges	priority_score	cluster_name	Contract	tenure
0	5760-IFJOZ	0.905784	107.95	97.779420	Vulnerable New Entrants	Month-to-month	3
1	6734-GMPVK	0.912003	105.30	96.033935	Vulnerable New Entrants	Month-to-month	5
2	4566-NECEV	0.920857	96.55	88.908783	Vulnerable New Entrants	Month-to-month	5
3	4060-LDNLU	0.922419	96.20	88.736721	Vulnerable New Entrants	Month-to-month	7
4	4445-ZJNMU	0.884382	99.30	87.819145	Uncommitted Mid-Tier	Month-to-month	9
5	2984-AFWNC	0.915583	95.40	87.346580	Vulnerable New Entrants	Month-to-month	3
6	6350-XFYGW	0.920468	94.75	87.214371	Vulnerable New Entrants	Month-to-month	4
7	9094-AZPHK	0.869586	100.15	87.088992	Uncommitted Mid-Tier	Month-to-month	15
8	7398-SKNQZ	0.914737	94.90	86.808517	Vulnerable New Entrants	Month-to-month	3
9	5150-ITWWB	0.914149	94.85	86.707026	Vulnerable New Entrants	Month-to-month	3
10	8161-QYMTT	0.918196	94.10	86.402278	Vulnerable New Entrants	Month-to-month	7
11	3292-PBZEJ	0.773851	111.40	86.207055	Uncommitted Mid-Tier	Month-to-month	11
12	4929-XIHVV	0.899874	95.50	85.937922	Vulnerable New Entrants	Month-to-month	2
13	2081-VEYEH	0.793593	107.95	85.668409	Uncommitted Mid-Tier	Month-to-month	3
14	5956-YHHRX	0.819223	104.35	85.485920	Uncommitted Mid-Tier	Month-to-month	21
15	9290-SHCMB	0.880608	96.85	85.286870	Uncommitted Mid-Tier	Month-to-month	13
16	8313-NDOIA	0.815876	104.15	84.973482	Uncommitted Mid-Tier	Month-to-month	24
17	9547-ITEFG	0.825364	102.60	84.682325	Vulnerable New Entrants	Month-to-month	9
18	5995-SNNEW	0.808857	104.45	84.485095	Uncommitted Mid-Tier	Month-to-month	23
19	0365-BZUWY	0.822868	102.55	84.385134	Uncommitted Mid-Tier	Month-to-month	17

Priority Score = Churn Probability × Monthly Spend. High churn probability on a low-spend customer scores lower than moderate probability on a high-spend customer. **This ensures the retention team focuses where revenue impact is highest.**

*Full priority breakdown - for all non churned segment

	customers	avg_score	avg_monthly	rev_at_risk
tier				
Low	1291	0.04	32.11	1518.88
Medium	1291	0.19	54.37	9395.75
High	1290	0.36	71.18	28923.37
Critical	1291	0.62	87.58	68996.82

If retention capacity is limited this quarter, outreach should be prioritized toward customers in the Critical tier, where revenue risk is highest.

Behind the Analysis

Detailed EDA workflow, clustering approach, and technical implementation can be reviewed in the supporting Google Collab Notebook [EDA Telco Customer Retain Analysis](#)

EDA Telco Customer Retain Analysis.ipynb

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EDA Telco Customer Segmentation & Retention Analysis

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- Date Created : 22-May-2026

General Context

Hi Guys 🙋🏻, this is my personal portfolio project to learn and exercise my skills in doing exploratory data analysis for a telco company's customers, specifically for segmentation & retention.

This notebook walks through the **full end-to-end process** — from raw data to actionable retention strategy. Here is what is covered:

- Step 1 — Data Quality Assessment.** Validated 7,043 records, fixed data types, handled missing values in TotalCharges, and confirmed the data was clean before analysis began.
- Step 2 — Exploratory Data Analysis.** Broke down churn patterns across demographics, services, contracts, and payment methods. Identified which customer segments carry the highest and lowest risk before building any models.
 - Layer 1 — Rule-Based Persona.** Designed 3 customer personas using business logic and domain knowledge. This established the baseline segmentation and revealed where manual rules hit their ceiling.
 - Layer 2 — Statistical Clustering (K-Prototype).** Let the data define natural customer groups. Used elbow method and silhouette score to select K=5. The clusters revealed behavioral segments that manual rules could not capture.
 - Feature Validation — Mutual Information & XGBoost.** Identified which features actually predict churn. Confirmed 4 key drivers that all three predictive models agree on: tenure, contract type, support/security add-ons, and service zone.
 - Layer 3 — Predictive Scoring.** Built and compared Logistic Regression, Random Forest, and XGBoost. Selected Random Forest as the best model (AUC 0.832). Scored every active customer by churn probability, ranked them by revenue at risk, and produced a ready-to-use retention priority list.
- Step 3 — Stakeholder Deliverable.** Translated all findings into a presentation deck that tells the full story for business stakeholders — from customer segments and churn drivers to the retention priority list. The goal was not just to build models, but to make the insights actionable for stakeholders who need the business takeaway, not the technical details.

If you are here to see how customer segmentation and churn prediction work in practice from cleaning data to delivering a prioritized action list, this notebook walks through every decision, trade-off, and business interpretation along the way.

[Main Questions to Answer]

Strategy to Retain Customers

Hi there!



Thank you for reading through! I'd love to hear your thoughts.

If you have any questions, feedback, or ideas on how to improve this, please feel free to share. ✨

Why is this documentation (Notebook & Deck) structured this way? A few reasons:

- **For Personal Growth:** It is a space for me to practice, refine, and share my end-to-end EDA workflow.
- **For the Community:** Around 4 years ago, when I faced my first EDA project, I drew a total blank. I created 1001 charts and visuals, but I had no idea how to connect them into a meaningful story, and I didn't know where to start. My mentor at the time just told me it was all about experiences, which felt incredibly ambiguous. Long story short, while I am still learning and improving today, my framework is much clearer now than it was back then. I hope sharing this workflow can help you in any way.
- **For Future Me:** Let us be real. In three months, I won't remember half of what I did here. This write-up ensures that I can easily understand my own work when I look back at it later. 😊

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